

Weather, Monitors, or Policy? Results summary

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3 October 2026

Question. Once weather and changes in the monitoring network are removed, did PM2.5 change differently in cities enrolled in India's National Clean Air Programme (NCAP) than in comparable cities? I registered the analysis plan before comparing NCAP with non-NCAP cities after 2018 (<https://osf.io/jksne/>). Full report, code and decision log: <https://github.com/breenu/ncap-evaluation>.

RQ1: measurement (ground monitors, descriptive).

- In the 18 NCAP cities with a PM2.5 station valid every year from 2018 to 2025 (16 with a single such station), the reported change of -23.6% becomes -14.1% after removing modelled weather and the change in which monitors exist. **H4 supported:** $+9.5$ pp (95% CI $+4.8$ to $+14.5$), of which network composition is $+5.8$ pp. For PM10 (13 cities): $+6.0$ pp ($+1.8$ to $+10.0$), mostly modelled weather. H4 holds in 31 of 31 PM2.5 versions and 29 of 31 PM10 versions.
- New stations read cleaner than existing ones in the same city and year: -6.3% raw, -6.8% deweathered (PM2.5). Across 23 cities with a 2018 panel, network composition makes the average city look 3.7 pp better by 2025 (-7.2 to -0.4).
- The audit found instrument ceilings, flatlines and 653 level shifts; missing days are *less* common on polluted days, so naive annual means are biased slightly upward ($+0.7\%$).

RQ2: weather. Weather moves a city's annual PM2.5 by a median 5.1% from one year to the next (12.6% at the 90th percentile), against a median raw year-on-year change of 10.3%.

RQ3: the satellite comparison (confirmatory). 113 urban centres containing 121 NCAP cities against 923 comparable non-NCAP centres, 2010–2024, synthetic difference-in-differences.

- Satellite PM2.5 fell in both groups after 2018: -16.8% in NCAP units, -24.3% in the comparison pool.
- **H1: not identified by this design (failed: (b) pre-trend Wald test).** NCAP units' satellite PM2.5 did not fall relative to comparable units; the estimates point to a relative rise of about 3–5%. H1 is not identified by this design, so this is not an effect of NCAP. Primary estimate $+3.6\%$ (95% CI $+2.4\%$ to $+4.8\%$); pre-trend Wald $p = 0.0008$; 17 of 19 registered checks agree, every one positive. H2 was not tested.
- Calibration leakage: the registered warning fires (gained a monitor $+2.5\%$, did not $+5.5\%$). Raw MAIAC AOD, which no monitor calibrates: $+0.8\%$ (-0.4% to $+2.0\%$) against $+4.2\%$ for satellite PM2.5 on the same units ("rise not reproduced in AOD"); the monitor-gain gap is absent ("gap absent from AOD (consistent with calibration leakage)"), though the two signals diverge most where no monitor was added (exploratory). AOD is read for direction only.
- Ground layer (secondary, one pre-year): PM2.5 before–after -14.0% , against control cities -4.7% (-13.4% to $+4.7\%$). Ground and satellite agree that these cities got cleaner; the satellite says not more than comparable cities.

RQ4: heterogeneity and mechanism (city-level relative changes, not effects of NCAP). City estimates are too noisy to rank: a typical 95% rank interval spans 75 of 113 places, and 0 pass Benjamini–Hochberg. **H5** (smaller changes in the Indo-Gangetic Plain): registered rule not met (-0.3% , -3.3% to $+2.6\%$). **H3** (dust control): inconclusive. Funding dose (exploratory): -0.4% per doubling of allocation (-3.5% to $+2.7\%$).

Key caveats. H1 is not identified by this design, so no number here is an effect of NCAP. NCAP cities are larger and less often in the Indo-Gangetic Plain than the comparison pool. Satellite PM2.5 is calibrated to ground monitors. Ground results rest mostly on single monitors and one pre-NCAP year. The confirmatory layer measures PM2.5, while NCAP's targets are for PM10. Every deviation from the registered plan is listed in Appendix A of the report.